Course Code:
Course Title: Big Data Analytics & Mining
Credits: 3.0
Rationale of the Course: This course introduces the concepts, tools, and techniques required to process, analyze, and extract knowledge from large-scale datasets. With the rapid growth of data in various domains, traditional data processing methods are no longer sufficient. This course focuses on distributed computing frameworks, data mining techniques, and scalable analytics solutions. It also explores real-world applications such as business intelligence, social network analysis, and machine learning on big data platforms.

Course Content: Introduction to Big Data: Characteristics (Volume, Velocity, Variety, Veracity), Challenges and Opportunities, Big Data Ecosystem and Technologies, Distributed Computing Fundamentals; **Hadoop Framework:** HDFS Architecture, Map Reduce Programming Model; **Spark Framework:** RDDs, Data Frames, Spark SQL; **Data Storage Systems:** NoSQL Databases (Key-Value, Document, Column-Family, Graph Databases); **Data Preprocessing:** Cleaning, Integration, Transformation, and Reduction; Data Warehousing and ETL Processes; Data Mining Concepts and Techniques; **Association Rule Mining:** Apriori, FP-Growth algorithms; **Classification Techniques**: Decision Trees, Naïve Bayes, k-NN; **Clustering Techniques:** k-Means, Hierarchical Clustering, DBSCAN; Regression Analysis and Predictive Modeling; Stream Data Processing and Real-Time Analytics; Text Mining and Sentiment Analysis; Social Network Analysis; Visualization Techniques for Big Data; Big Data Security, Privacy, and Ethical Issues; **Applications of Big Data in Industry:** Healthcare, Finance, E-commerce, Smart Cities; Case Studies and Emerging Trends in Big Data Analytics

Course Learning Outcomes (CLOs): At the end of the course, student will be able to-

CLOs	Statements
CLO1	Understand the fundamental concepts, architectures, and challenges of big data systems.
CLO2	Apply distributed computing frameworks for large-scale data processing.
CLO3	Analyze datasets using data mining and machine learning techniques to extract meaningful insights.
CLO4	Evaluate and apply big data solutions in real-world scenarios, considering ethical and security issues.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√									
CLO4		√		√								

Mapping of Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture, Multimedia	Written Examination
CLO2	Lecture, Multimedia	Written Examination
CLO3	Lecture, Problem Solving	Written Examination
CLO4	Lecture, Group discussion	Written Examination

Textbooks:

- Marz, N., & Warren, J. (2015). *Big Data: Principles and Best Practices of Scalable Realtime Data Systems*. Manning Publications.

Reference Books:

- Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of Massive Datasets* (3rd ed.). Cambridge University Press.
- White, T. (2015). *Hadoop: The Definitive Guide* (4th ed.). O'Reilly Media.
- Chambers, B., & Zaharia, M. (2018). *Spark: The Definitive Guide: Big Data Processing Made Simple*. O'Reilly Media.